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Contracting, market access and deforestation ☆

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ABSTRACT

We study the impacts of market access on forest loss in Ghana through a program designed to increase smallholder participation in oil palm commodity markets. Improved market access is facilitated through production contracts in which smallholders receive credit to establish production, a guaranteed price and quantity for the contract duration, and output pickup at the village. Using a variety of difference-in-differences approaches, we find substantial increases in forest loss in targeted villages following the introduction of the contracting program. The findings suggest that the ecological impacts of reforms to market access may be sizeable.

1. Introduction

Rural households in developing countries often face high barriers to market participation that prevent structural transformation in agriculture and limit the incomes of smallholder farmers. In regions of the world where much of the local population is dedicated to agriculture, barriers to market participation—particularly limited credit, high transportation costs, and high transaction costs—impede growth and lower welfare. The theoretical and empirical evidence largely indicates that improving market access for rural households promotes growth and improves welfare in a variety of settings (e.g. Breza and Kinnan, 2021; Burgess and Pande, 2005; Sotelo, 2020; Gollin and Rogerson, 2014).

Despite the potential economic benefits, improved market access may have negative impacts on the local environment. Market access might change the relative value of land in agriculture or facilitate credit access that leads to agricultural extensification at the expense of natural land. Likewise, income growth may increase demand for land-intensive goods. Externalities from land conversion to agriculture include biodiversity loss, increased global greenhouse gas emissions, local soil degradation, and increased exposure to climatic shocks. Such external costs may be higher in developing countries in the tropics due to the carbon density in forests, endemic biodiversity, and limited resources to protect against soil erosion or increased health risks. Economic gains from improved market access may be partially undermined if environmental degradation follows these reforms. On the other hand, market access might allow smallholders to utilize more inputs to improve yields and intensify production—an effect which in and of itself has *ex ante* ambiguous predictions for the local environment (Byerlee et al., 2014)—or may lead to greater demand for environmental protection and sustainable agricultural practices. The potential ambiguity of market access on the environment requires careful, well-identified empirical work to disentangle complicated transmission channels.

In this paper, we offer plausibly causal evidence on the impacts of improved market access on forest loss through a program designed to reduce barriers to entry in oil palm production for selected villages in Central Ghana. In our setting, market access is facilitated through production contracts with a large palm oil mill in the region which made it easier for farmers to begin growing oil palm and lowered barriers to marketing oil palm output. The contracts eliminated investment constraints by extending credit to cover the extremely high fixed costs of establishing oil palm plots. This credit arrangement allowed farmers to borrow and cover the high startup costs where they otherwise had little or no access to credit and few alternatives to begin commercial

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production. The program also eased market access barriers by dramatically reducing transportation and transaction costs. The contracts fully enumerate transactions in advance—including buyers, sellers, prices, quantities and, critically, pickup location at the village—in a setting in which search frictions and transportation costs are otherwise high. Notably, this program was established in villages where farmers faced clear barriers to credit and oil palm market participation.

We link a variety of geospatial data to local Ghanaian villages, including high-resolution annual estimates of forest loss from 2001 to 2019. We compare changes in forest loss surrounding villages that participated in the contracting program in 2008 to those that did not via a variety of difference-in-differences approaches (difference-in-differences, matched difference-in-differences, and synthetic difference-in-differences). We find substantial increases in forest loss in targeted villages immediately following the program roll-out that coincides with the timing of farmer enrollment in the program. Furthermore, we find that higher average annual forest loss among participating villages continued over the following decade—evidence consistent with persistence in the program effect. These results are robust to different control group specifications. Fully specified event studies provide strong evidence against spurious findings driven by pre-existing differences in forest loss trends.

The primary contributions of this paper are twofold. First, we present well-identified evidence on the relationship between several dimensions of market access and the environment. Causal identification of market access is challenging, as variation in access is often driven by large, macroeconomic changes, country-level governance institutions, regional investments in infrastructure, or even a local community's ability to solve coordination problems and engage in collective action (Markelova et al., 2009). These drivers of market access also have their own direct implications for land use change and forest loss that prevent causal inference. Our setting allows us to study the ecological impacts of a program that improved market access in isolation from other economic changes that would otherwise confound the analysis. Our second major contribution identifies specific, partial equilibrium transmission channels of market access shocks. Critically, the program we study is narrow in scope—only affecting oil palm production—which allows us to identify causal channels isolated from broader general equilibrium effects. For example, road construction reduces transportation costs and increases market access, but does so for nearly all markets at once (e.g. Asher and Novosad, 2020; Brooks and Donovan, 2020). Likewise, access to broad credit markets can affect rural economies through a variety of channels, including aggregate demand and business investment (e.g. Breza and Kinnan, 2021). While quantifying general equilibrium effects is a critically important task for researchers, so too is measuring partial equilibrium effects along particular transmission channels. While the program shock we study exhibits several complementary features, the channels are considerably narrower in scope than previous market access shocks used in the literature. The variation in market access is only for a single agricultural commodity—palm oil.

The institutional features of the studied program also relate to several broader themes in the literature. First, the introduction of credit for producers speaks to the broader literature on the role of credit and land use. Credit market access has important implications for development and agricultural decision making. Generally, credit access improves welfare in rural communities (Burgess and Pande, 2005; Breza and Kinnan, 2021). In the context of deforestation, it is unclear whether access to credit increases or decreases forest loss. Harvesting of forest products may be a means to smooth consumption for smallholders who face liquidity constraints or negative shocks to agricultural productivity (Noack et al., 2019; Jayachandran, 2013). In these cases, access to credit markets may offer smallholders alternative consumption smoothing opportunities. However, if credit allows landowners to expand agriculture or pasture, then they may choose to clear more land under credit and use the increased production to make payments. Indeed, Assunção et al. (2020) find that credit constraints imposed on cattle ranchers in Brazil substantially reduced deforestation. We bring additional evidence to bear on this question in a new smallholder setting focused on credit-constrained agricultural investment.

Second, the program organized output product pick-up from village centers which dramatically reduced the barriers to marketing via reductions in transportation cost risks faced by producers. Reductions in transportation costs can have large impacts on local economic development, production decisions, and agricultural profits (Gollin and Rogerson, 2014; Datta, 2012; Platteau, 1996). Empirical studies on these relationships typically examine the impacts of infrastructure improvements on the outcomes of interest (Sotelo, 2020; Brooks and Donovan, 2020). Infrastructure improvements may reduce the transportation costs to get goods to market, but may also connect rural areas to urban labor markets, reduce costs of purchased inputs, and offer a variety of other general equilibrium effects (Asher and Novosad, 2020; Brooks and Donovan, 2020). Much of the early empirical work on roads and deforestation finds blanket decreases in transportation costs for all goods lead to increased forest loss—e.g. Pfaff (1999), Pfaff et al. (2007)—however, new-road induced reductions in transportation costs reduce costs for both agricultural goods and timber which effectively decreases clearing costs while also increasing net returns for agricultural products. Recent work, e.g. Asher et al. (2020), Baehr et al. (2021), demonstrates that the relationship between new road construction and deforestation may be more nuanced. Because the reduction in transportation costs in our setting applies only to a single output product, we can better isolate the impacts of removing this particular barrier to market access and thus ensure that the effects we find are not driven by changing labor movements or reductions in clearing costs.

Finally, in our empirical setting (as in many other contracting schemes) the contracts reduce search costs by linking sellers and buyers for the output before the goods are harvested and also reduce the output price risk faced by the producer. The widely-held belief that reductions in transactions costs improve smallholder income and increase market participation has considerable support in the literature (Chamberlin and Jayne, 2013; Bellemare and Barrett, 2006; Renkow et al., 2004; Key et al., 2000).[2] There is a large literature on contract farming, but a common feature in this work is that causal inference on the impacts of contract farming is often difficult due to farmer selection into contracting (Barrett et al., 2012; Bellemare and Bloem, 2018). The village-level nature of the contracting program and timing of its introduction allows us to credibly identify village-level environmental impacts sidestepping concerns about selection at the farmer level. The other key feature of these contracts is the reduction in output price risk (Bellemare et al., 2021). Price risk has important implications for agricultural decision-making (see Boyd and Bellemare (2020) for a recent review of this literature) and thus implications for natural land conversion to agriculture. Lundberg and Abman (2022) finds reductions in maize price volatility increase forest loss and maize production across sub-Saharan Africa. The contracting scheme we study speaks to the broader literature on transactions costs and producer price risk.

The remainder of the paper is as follows. We begin by providing background on oil palm farming in Ghana and details on the oil palm contract program we study. We then describe the data sources we use for the analysis and outline our empirical strategy while providing evidence for our identifying assumptions. We present our findings across a variety of specifications, describe the additional analysis presented in the supplemental materials and conclude with some final remarks.

[2] Meemken and Bellemare (2020) find increases in household income from contracting but limited evidence of broader development effects via hired labor.

2. Background

Oil palms are native to Ghana with a long history of smallholder cultivation. Smallholder farmers produce about 60% of palm oil output in Ghana and account for 85% of the planted area (Khatun et al., 2020). Despite familiarity with palm oil as a commodity crop, many Ghanaian smallholders have failed to take advantage of the global boom in palm oil demand in large part due to barriers to credit markets (Khatun et al., 2020). The customary land tenure system governing much of the agricultural land in Ghana recognizes individual ownership of standing crops, but not for the soil below (Pande and Udry, 2005). Without transfer rights, the user of the land may not put up their plot as collateral for agricultural credit. Although expropriation risk to standing crops is low in Ghana, the inability to use the land as collateral for loans likely discourages productive investment—particularly in perennial crops that require a number of years before generating revenue. Improving access is likely to encourage farmers to participate in the oil palm market, but may also drive local land use change at the expense of sensitive ecosystems as oil palm monoculture displaces native forest.[3] Oil palm production, particularly in Southeast Asia, is strongly associated with large-scale forest loss (see e.g. Busch et al., 2015; Edwards, 2019; Cisneros et al., 2021; Kraus et al., 2021).

Our quasi-experimental identifying variation comes from a palm oil contracting program introduced in 2008 by the Twifo Oil Palm Plantation (TOPP) in the Central Region of Ghana. TOPP is a large palm oil plantation with a high-capacity palm oil mill. Despite growing very large quantities of oil palm fruit on the plantation, the processing capacity of the mill is high enough that TOPP is unable to fully meet mill capacity with their own output. As a result, TOPP developed a resource-providing contract program to secure additional oil palm output from smallholder farmers in the region. TOPP introduced the program to selected villages within their catchment and offered contracts to all households in the program villages.[4]

The program has three features that improve market access for treated villages. First, the contracts allow households the option to fund the establishment of oil palm plots solely on credit through TOPP. This credit allows smallholder farmers to amortize the considerable startup costs associated with oil palm production—average costs of smallholder palm plantations from establishment to first harvest are approximately 1,600 US dollars per hectare (Ruml and Qaim, 2020). With average plot sizes of approximately 2.5 hectares, these startup costs represent nearly 2.5 times the mean annual income for rural Ghanaian households (Ghana Statistical Service, 2008). Without access to credit, smallholders may be prevented from making startup investments in oil palm plots even if doing so has a positive expected net present value. Additionally, the contracts alleviate access barriers to oil palm inputs more broadly. Participating households can purchase inputs on credit, including planting materials, tools, machinery, and chemical inputs. Participating households can establish the plot themselves or elect for TOPP staff to establish the oil palm plot entirely on credit (including labor costs of preparation and planting). Farmers that utilize credit via the contract are charged an annual interest rate of 11.5% with in-kind repayment through future oil palm output. Smallholders only receive payment for three quarters of their output as the remaining one quarter is used to pay down any existing credit balance until paid in full. Loans are secured by the oil palm plot and the contract allows TOPP to take control of the plot if farmers stop delivery of oil palm fruit before loan repayment is complete. This credit access is available throughout the contract duration. This feature of the program represents a persistent, rather than transitory, shock to credit constraints that may lead to persistent ecological impacts. This contracting arrangement is notably different than many other production contracts that specify particular production methods and input quantities—TOPP contract participants maintain agency over production decisions. Indeed, TOPP does not provide guidance or training on the use of particular inputs and technical assistance more broadly is limited to plot establishment by TOPP staff. Second, TOPP arranges transportation to pick up oil palm output in treated villages, significantly reducing prohibitively high marketing and transportation costs for remote rural smallholders. This effectively brings the commodity market to the smallholder and allows them to participate in palm oil commodity markets that they otherwise would not have access to. Notably, contract households are prohibited under the contract from selling palm oil output to any other buyers. Hence, the contract uniquely establishes the output market location for the entire oil palm output. Finally, like many agricultural contract regimes, the TOPP program reduces output price risk for smallholder farmers by establishing fixed prices for oil palm output thereby reducing transaction costs associated with marketing oil palm output. Contract prices are fixed for the year but are adjusted year-to-year based on price changes in international palm oil markets.

3. Data

We use a variety of different data sources in the analysis. For information on participating villages in the contract program, we use data from Ruml and Qaim (2020) and Ruml et al. (2021).[5] These data include geo-coordinates for surveyed villages and allow us to identify the locations of 13 different villages that participated in the contract program. The data also include household- and plot-level survey response data for a sample of households participating in the program. Survey villages and households were selected randomly from participating households. The survey was conducted in 2018 but asked a number of retrospective questions regarding contracting, previous experience with oil palm production, and previous land use of contracted oil palm plots. Through contacts in Ghana, we were able to locate an additional 4 non-surveyed villages that participated in the contracting program.[6]

The household and plot level data allow us to confirm underlying assumptions regarding oil palm expansion in response to the contracting program. First, we see that contracting participation increases rapidly around the 2008 start of the program with more than 80% of all surveyed households reporting their first oil palm contract was between 2008 and 2009 with small subsequent increases in 2010 and beyond (see Figure A.1). Second, the vast majority of surveyed plots (84%) were converted to oil palm after the farmer entered into a contract. One fifth of these plots were forest land prior to conversion and the plots are sizeable—approximately 6.25 acres on average.[7],[8] Finally, smallholder survey respondents clearly faced palm oil market access frictions—only 16% of surveyed participant households reported growing oil palm prior to the introduction of the market access program. Survey responses indicate that credit access and high startup costs are clear barriers to market entry. No survey households reported having any credit access when the contracting program was introduced. Furthermore, more than 90% of surveyed households elected for TOPP staff to

[3] Koh and Wilcove (2008) find that land conversion for oil palm in Indonesia and Malaysia significantly impacted biodiversity.

[4] Program details and data come from Ruml and Qaim (2020), Ruml et al. (2021), and correspondence with authors.

[5] The data set was graciously provided by the authors.

[6] We located a total of 9 additional villages, however 5 of these villages joined the program later and we were unable to confirm the year that they joined and thus omit them from the analysis. We obtained this information from TOPP staff via a US Department of Agriculture Foreign Agricultural Service intermediary posted to the US Embassy in Ghana.

[7] Tree cover loss—as measured in this paper—may also arise from pasture or agricultural land conversion so long as standing trees exist in the area

[8] We present the kernel density estimates of oil palm plot size by previous land use in the Appendix. Plots of previously forested land are slightly larger than those previously in pasture or agriculture/crops.

establish oil palm plots on credit. The important limitation of the survey data is that it does not identify non-participating households or villages in the surrounding area and, thus, limits our ability to compare plot or household level differences in outcomes in the observed, participating villages to a set of control villages. Additionally, the survey data do not include detailed contract-level information (e.g. contract duration, annual contract prices, etc.).

We identify a set of other villages in the surrounding area using the Geo-Referenced Infrastructure and Demographic Data for Development (GRID3) data set on settlement extents. This data set identifies the boundary of cities (referred to as built-up areas or BUAs) and the location of small settlement areas (SSAs) via satellite imagery. We use SSAs as a proxy for villages and allow the SSAs that do not overlap with those in the program data set to serve as our control villages. Visual inspection of satellite imagery, as well as consistency between surveyed village-level geo-coordinates and SSA observations, suggest the GRID3 SSAs are a suitable source for non-surveyed villages in this region.

The GRID3 data set provides the locations of villages throughout the country. To attribute land area to village locations (both program and control villages), we create Thiessen polygons (a.k.a Voronoi polygons) around SSA centroids. These polygons are commonly used in settings in which researchers have the universe of village locations but no formal, demarcated boundaries exist (e.g. Alix-Garcia et al. (2013), Heßet al. (2021)). To ensure that no urban land is attributed to a particular village, we remove any overlapping area of our Thiessen polygons with the extent of the BUAs. This ensures that all area attributed to a village falls outside of any urban or semi-urban area.

With both the coordinates of the villages and the boundaries of the Thiessen polygons, we can attribute several different spatially explicit characteristics for each village. We construct average elevation and ruggedness for each village at the polygon level. Elevation is the average elevation of all land within the polygon and ruggedness is the standard deviation of elevation in the same area. We collect historical agro-climatic estimates of potential oil palm yield under rain-fed, high-input production using the Global Agro-Ecological Zoning (GAEZ) dataset (version 4.0). We average the estimated yields within Thiessen polygons to obtain average palm oil potential (converted to tonnes of oil/acre for Table 1). We calculate Euclidean distance to the nearest market center utilizing the market locations from Porteous (2019). We also calculate the distance from the village location to the nearest primary road and nearest secondary road using spatially-explicit road network data from OpenStreetMap via the World Food Programme's GeoNode data portal. The final distance measures we calculate are the distance between the village center and the Twifo oil palm processing facility at the Twifo Oil Palm Plantation.

Our data on tree cover and forest loss come from the Global Forest Change dataset (Hansen et al., 2013). The GFC dataset provides high-resolution, spatially-explicit estimates of treecover and annual forest loss across the terrestrial surface of the earth. Version 1.7, the version used in the analysis for this paper, provides annual estimates of loss for 2001–2019. We aggregate the 900-square-meter pixels to the village-level using our Thiessen polygons and attribute any forest loss that lies within a particular polygon to the nearby village. We require that pixels have at least 10% of their area in tree cover to be classified as forest land at baseline. Forest loss in pixels with less than 10% is not counted in our analysis. Our 10% threshold for tree cover liberally captures sparser tree canopy cover common throughout much of sub-Saharan Africa and our results are robust to much more stringent thresholds of 30% and 50% which we report in the Appendix. The GFC dataset only records forest loss on land that was classified as having treecover at baseline. This means any growth that occurred after 2000 and may have been subsequently cut down will not be recorded as forest loss which avoids measures of forest cover or loss being affected by growing oil palms planted during the study period.

The data processing described above yields a final dataset comprised of 930 unique villages in the Central Region of Ghana, of which 17 we know to be participants in the Twifo contracting program beginning in 2008. We conduct our main analysis by limiting the sample to villages in the Central Region to account for common institutional features that vary across regions in Ghana. In particular, there are a variety of governance characteristics and administrative functions at the region level in Ghana that affect policy implementation and rural development (e.g. Abdulai and Hulme, 2015; Ayee, 1997).[9] We note, however, that results presented in the Appendix using all villages within 100 km of the Twifo village—regardless of region—look very similar in magnitudes and significance across all main specifications. Table 1 presents summary statistics for each of these measures for contracting villages, matched control villages, and all other control villages in the Central Region. We present maps of baseline forest cover, average rate of forest loss prior to 2008, and our village classification in Figure A.3 in the supplementary analysis. Fig. 1 presents the average annual rate of forest loss from 2001 to 2019 for three groups of villages; villages participating in the resource contracting program, all other villages in Ghana's central district, and the set of matched control villages. Program effects are visible as elevated forest loss in contract villages relative to control villages following program implementation. Fig. 1 also reflects fairly high variation in annual deforestation rates in the region that we speculate may be related to movements in palm oil prices. While we do not have local Ghanaian price data for oil palm fruit or palm oil that might allow us to confirm this hypothesis, contract program prices are referenced to international prices and these patterns of deforestation appear at least partially related to broad patterns in international palm oil prices. Notably, we see similar patterns in deforestation from 2014 onwards in all villages in the region, suggesting that there were regional deforestation pressures—market prices or otherwise—at play during this period.

4. Empirical approach

We estimate the impact of the contracting program on forest cover in the program villages using the program introduction as an exogenous shock to market access. Our identification strategy leverages the panel nature of the dataset to estimate a difference-in-differences model:

$$Defor_{it} = \beta_1 Contract_i \times Post2008_t + \alpha_i + \gamma_t + \epsilon_{it} \tag{1}$$

where $Defor_{it}$ is an annual measure of deforestation—either the inverse hyperbolic sine (IHS) of hectares of treecover lost or the rate of forest loss—around village i in year t, $Contract_i$ is a binary treatment variable equal to 1 if village i receives market access via contracts, and $Post2008_t$ is a binary variable equal to 1 after 2008 when the contract program enters into effect. α_i and γ_t are village and year fixed effects which control for any time-invariant village characteristics that drive village forest loss (e.g. village population, agricultural suitability, etc.) and common shocks that affect all villages in our sample (e.g. oil palm prices, weather-related productivity shocks, etc.), respectively.

We estimate the above model using three different approaches. First, we use all untreated villages in the Central Region as our control group and estimate the model above. Second, we employ a matched-sample difference-in-differences strategy. This pre-processing approach performs better than full-sample difference-in-difference in certain settings (Ferraro and Miranda, 2017). For every treated village, we select four control villages that most closely resemble the treated village in question on the observable characteristics described above, specifically: share of tree cover at baseline area of the Thiessen polygon (using 10%, 30% and 50% thresholds), average elevation, ruggedness,

[9] For example, Regional Coordinating Councils exercise oversight and some degree of control in coordinating local district-level budgets, which include local development plans and programs. Likewise, national ministries also have region-level offices and staff that administer central government programs and policies (Ayee, 1997).

Table 1
Summary statistics for Ghana Villages by grouping.

	Village type		
	Contract (treated)	Matched control	Other untreated
Average forest loss (ha) before 2008	3.650	6.796	7.637
	(1.763)	(5.234)	(7.641)
Average rate of forest loss before 2008	0.00441	0.00593	0.00827
	(0.00149)	(0.00391)	(0.00609)
Share with > 10% treecover*	0.999	0.997	0.972
	(0.00223)	(0.00667)	(0.0715)
Total Forested Area (km^2)	9.173	11.12	9.455
	(5.924)	(5.820)	(7.431)
Area (km^2)*	9.186	11.16	9.683
	(5.938)	(5.829)	(7.492)
Average elevation (m)*	161.7	142.4	87.65
	(11.08)	(18.07)	(41.69)
Ruggedness*	10.59	10.63	9.026
	(4.170)	(4.360)	(5.346)
Ave oil palm potential (tonnes oil/acre)*	2.049	2.005	1.616
	(0.00489)	(0.0669)	(0.393)
Distance to nearest market (km)*	95.06	90.85	75.17
	(3.035)	(4.686)	(19.09)
Distance to primary road (km)*	16.27	19.53	19.21
	(4.344)	(8.041)	(16.74)
Distance to secondary road (km)*	2.368	2.397	1.778
	(1.834)	(2.249)	(2.164)
Share of land in reserve/PA*	0.0908	0.110	0.0793
	(0.217)	(0.234)	(0.237)
Dist to Twifo plant (km)*	38.69	39.20	60.14
	(4.385)	(9.418)	(29.40)
Number of villages	17	30	878

Notes: This table presents summary statistics across three different groups of villages; treated villages (those participating in the Twifo contracting program), a matched set of control villages in from the Central Region, and all other untreated villages in the Central Region. Means of variables are above with standard deviations below in parenthesis. Asterisks denote that the variable was used in the matched procedure.

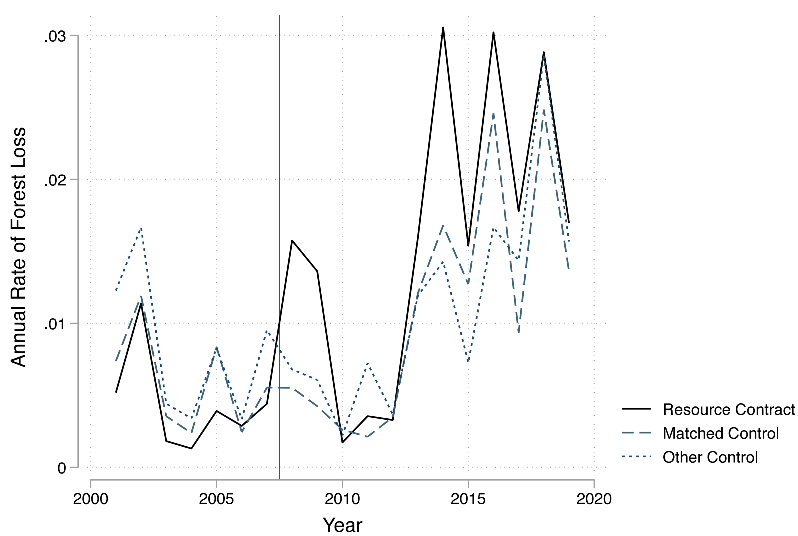

Fig. 1. Average annual rate of forest loss by group - Ghana.
Notes: The above figure presents average annual rates of forest loss over time for the treated villages (those participating in the Twifo resource contracting program), the matched non-participating villages, and all other non-participating villages in the Central Region. The vertical line denotes the start of the contracting program.

agro-climatic estimates of potential oil palm yield, distance to nearest market, distances to nearest primary and secondary roads, share of land in a game reserve or protected area, and distance to the Twifo plant.[10] We link treated villages to their nearest neighbors in attribute space by calculating the Mahalanobis distance between each treated village and every untreated village in the Central Region of Ghana.[11] We pair the treated village with the four untreated villages that are 'closest' based on this distance measure. We sample with replacement to avoid order bias or bias induced by matching to otherwise dissimilar

[10] We purposefully omit pre-treatment outcomes in our matching process to avoid potentially introducing bias in our estimates via mean-reversion as argued by Daw and Hatfield (2018). Furthermore, Ferraro and Miranda (2017) demonstrate that time-invariant characteristics can be more important in the matching process than time-varying, pre-treatment variables.

[11] The Mahalanobis distance between a treated village t and a control village c is given by $d(t,c) = \sqrt{(X_t - X_c)' \Sigma_{tc}^{-1} (X_t - X_c)}$ where X_t, X_c are vectors of attributes for the treated village and control village, respectively, and Σ_{tc} the variance–covariance matrix between these vectors.

villages.[12] We then estimate our main difference-in-difference model using the matched villages as the control group. Our final approach estimates the above model using the synthetic differences-in-differences approach developed by Arkhangelsky et al. (2021). This approach uses a weighted average of control villages to serve as the counterfactual for the treated villages and offers potential improvements over standard difference-in-differences models.

The identifying assumption underlying our analysis is that, absent the program, trends in forest loss observed in the 17 villages participating in the program would have continued in parallel to trends in forest loss in our control villages after 2008. While fundamentally untestable, Fig. 1 lends support for this assumption and indicates no systematic divergence in trends before 2008 among either the matched control group or the other 878 Central-Region villages. Furthermore, we estimate event-study specifications for our difference-in-differences and matched sample difference-in-differences models to formally test the presence of diverging pre-trends. Our synthetic difference-in-differences model yields parallel pre-trends in the counterfactual by construction and we present these plots in Figure A.4 in the supplementary materials. Under our identifying assumptions, β_1 in (1) will identify the average treatment effect on the treated (ATT) of the village-level program.[13]

In our difference-in-differences and matched difference-in-differences models, we calculate standard errors that allow for across-village correlation up to 5 km following Hsiang (2010). By allowing spatial dependence in the error term, we adjust inference to account for location-specific shocks that might affect multiple villages at the same time without assuming independence across villages and account for the potential of spatial mismeasurement that might arise via the use of Thiessen polygons to attribute land to a particular village. An alternative would be to cluster at the district level to allow for arbitrary within-district cross-correlation of error terms, however, the small number of districts observed in the Central Region prevents us from obtaining consistent estimates of standard errors and is further complicated by the fact that all treated villages lie in the same district. We note that clustering at the village level yields very similar standard errors as does increasing the distance kernel up to 25 km.[14] For our synthetic difference-in-differences models, we estimate standard errors via the clustered jackknife approach outlined in Arkhangelsky et al. (2021) as it is generally the most conservative across the possible approaches.

5. Results

We present our difference in differences estimates of Eq. (1) in Table 2. Panels A and B correspond to different outcome measures; panel A presents estimates using the inverse hyperbolic sine (IHS) of annual forest loss and panel B presents estimates using the rate of forest loss (i.e. annual loss divided by baseline forest cover). Columns (1) and (2) present results from the full-sample difference-in-difference models, Columns (3) and (4) present estimates from the matched-sample difference-in-difference models, and Column (5) presents estimates from the synthetic difference-in-differences models. All models include village and year fixed effects, while columns (2) and (4) add district time trends. Our coefficient estimates in the first panel imply an average annual increase in forest loss between 43% and 130%.[15] Comparing these estimates to the pre-treatment average forest loss among the treated villages, these estimates correspond to an average additional loss of 1.6 to 4.7 ha of forest area each year per village as a result of the program. Estimates in panel B are consistent with those above. They indicate large increases in the rate of forest loss for villages participating in the program. The increases range from 0.5% to 0.9% (relative to a pre-2008 mean rate of forest loss equal to 0.4%). All estimates are statistically significant at the one-percent level.

We present results from our event studies graphically in Fig. 2. Panels (a) and (c) correspond to models with the inverse hyperbolic sine of forest loss as outcomes and panels (b) and (d) correspond to models using the rate of forest loss as outcomes. Panels (a) and (b) present estimates using the full sample and panels (c) and (d) present estimates using the matched sample. All models include village and year fixed effects and include separate indicators for treated villages for each year from 2001 to 2019. We omit the year prior to the program (2007) to serve as our reference year, so all coefficient magnitudes can be interpreted as the change in the difference between treated and control groups relative to the 2007 difference.

Three notable patterns consistently emerge in all coefficient plots. First, there is no evidence of diverging trends in leading coefficients—i.e. those year-by-year estimates display no trends, with the majority not statistically differentiable from zero. This provides additional support for the parallel trends assumption that underlies our identification. Second, across all models, there is a dramatic increase immediately following the program. This increase coincides with the uptake in contracting (presented in Figure A.1 in the supplementary materials) and likely captures the initial clearing for oil palm crops following the start of the program. The third pattern is a persistent rise in program village forest loss relative to control villages from 2013 through the end of the sample period. This increase is somewhat muted in the matched sample analysis compared to the full sample but is still present and persistent through 2019.[16] This rise provides nuanced evidence of persistence in program treatment effects that we believe are related to the relaxation of liquidity constraints. The timing of this later increase in forest loss corresponds with the likely timing of the first harvest from oil palm planted following the 2008 contracting program as it typically takes three years for newly planted oil palm to reach maturity. Survey data also indicates an increase in farmers applying for other sources of credit around this time, many of whom report using the Twifo contract as collateral. We also note increases in control village forest loss during this period as illustrated in Fig. 1 and speculate that movements in international palm oil prices may have created deforestation pressures throughout the region. Program villages, however, with improved access to credit and with contract prices benchmarked to these international prices, may have been comparatively more responsive to these commodity market shocks due to improved access. Nevertheless, this evidence is merely suggestive without the data necessary to pin down mechanisms responsible for this apparent persistence in village-level treatment effects. All three patterns observed in the full-sample and matched sample event studies are consistent with differences in treated and synthetic control outcomes used in our synthetic difference-in-differences approach (presented in Figure A.4 in the supplementary materials).

We note that our main finding is robust to a variety of alternative specifications. First, results across all 10 specifications in Table 2 are very similar in magnitude and significance if we relax our restriction

[12] Because some control villages are matched to multiple treatment villages, we have 30 control villages in our matched panel analysis rather than 88.

[13] We are careful to point out that village-level treatment occurs at the same time. Variation in household participation illustrated in Figure A.1 arises from delays in household program uptake. If our unit of analysis was at the household level, β_1 would identify intent to treat (ITT) effects. Because treatment and our analysis are at the village level and treatment timing is the same for all villages, β_1 identifies the ATT.

[14] We present estimates of our main results with 25 km kernel distance in appendix Table A.7. The inference is largely unchanged, but estimates with district time trends in the full control sample are significant at the 5% level instead of the 1% level under 5 km spatial kernels.

[15] These effects are calculated from the following partial elasticity: $e^{\hat{\beta}} - 1$ as detailed in Bellemare and Wichman (2020). These elasticities themselves are also highly significant.

[16] In all cases, the average effect across the 2011–2019 period is both statistically and economically significant.

Table 2
Estimates of contracting market access program on forest loss.

Panel A - IHS annual forest loss					
	Full DiD		Matched DiD		Synth DiD
	(1)	(2)	(3)	(4)	(5)
Contract ×Post 2008	0.770***	0.357***	0.614***	0.468***	0.829***
	(0.122)	(0.138)	(0.0954)	(0.105)	(0.072)
District Time Trend		✓		✓	
Village FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Observations	17,575	17,575	988	988	–
R^2	0.004	0.041	0.055	0.078	–
Effect at mean (ha/yr)	4.2	1.6	3.1	2.2	4.7
Panel B - annual rate of forest loss					
	Full DiD		Matched DiD		Synth DiD
	(1)	(2)	(3)	(4)	(5)
Contract ×Post 2008	0.00869***	0.00490***	0.00575***	0.00503***	0.00710***
	(0.00145)	(0.00156)	(0.000927)	0.000970	(0.00117)
District Time Trend		✓		✓	
Village FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Observations	17,575	17,575	988	988	–
R^2	0.003	0.031	0.033	0.053	–
Effect relative to mean (%)	200	113	132	116	163

Notes: This table presents a variety of difference-in-differences estimates of the resource contract on forest loss from Eq. (1). Panel A presents coefficient estimates using the inverse hyperbolic sine of annual forest loss as the outcome variable while Panel B presents estimates using the rate of annual forest loss. Columns (1) and (2) present coefficient estimates from the standard, full-sample models. Columns (3) and (4) present estimates from the matched difference-in-differences models, and Column (6) presents estimates from the synthetic difference-in-differences model outlined in Arkhangelsky et al. (2021). All estimates include village and year fixed effects and Columns (2) and (4) add district time trends. Standard errors allow for spatial dependence up to 5 km in columns (1) – (4). Column (5) presents jackknife standard errors clustered at the village level. Statistical significance denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

that control villages can only come from the Central Region and instead allow any control villages to come from those within 100 km of the Twifo facility. We estimate our full-sample difference-in-differences specification using only the 13 surveyed villages in the treatment group and find estimates that are similar to those in our main model. The effect size is slightly larger and our precision actually increases slightly in models with the rate of forest loss. These results are presented in Table A.2. We ensure that results are not driven by systematic changes to all villages in close proximity to the Twifo mill in 2008 by adding an indicator for all villages within 50 km of the mill interacted with a post-2008 indicator term to our basic model. Were the increases in forest loss found in our main specification driven by proximity to the mill rather than the contracting program itself, this specification should reduce/remove the explanatory power of the contracting program. This is not the case as all β_1 estimates remain large and statistically significant. We present these estimates in Table A.3. Next we demonstrate that our findings largely hold for more stringent initial definitions of forest land in Table A.4. When we use a 30% or 50% threshold for defining forested land at baseline, our results are qualitatively similar, in fact magnitudes seem only to increase. This result indicates that our findings are not particular to land only with relatively sparse tree cover at baseline. Finally, we present results varying the control group by distance to the Twifo facility (50, 100, and 150 km) and region (Central Region vs any region) in Tables A.5 and A.6. Here too, our results remain remarkably stable indicating that our findings are not particular to a choice of a control group.

We also conduct a placebo exercise to examine the distribution of estimates that would arise by chance in our control villages. We approach this in two ways. First we randomly assign treatment to 17 villages from the Central region and estimate Eq. (1) (using the inverse hyperbolic sine of forest loss) as if those villages had participated in the Twifo contracting program in 2008—we refer to this as an IID sampling approach to reflect that placebo treatment villages are drawn from an independent and identical sampling distribution. In a second sampling procedure, we account for the spatially clustered pattern of treatment in the contracting program by randomly selecting a single village and assigning treatment to this village and the 16 closest villages—we refer to this as a clustered sampling approach. For each placebo sample, we estimate our main model specification, repeating this exercise 10,000 times for each sampling approach to construct a null distribution. Only 3% of our clustered sampling placebo estimates are larger in absolute value than the effect we observe in our main specification and none of the IID sampling placebo estimates were larger than the estimated effect size. The placebo exercise yielded t statistics larger than that observed in our main finding in only 1.1% of the IID sampling placebo estimates and in only 0.04% of the clustered sampling placebo. These distributions are presented in Figure A.5 in the Appendix.

6. Final remarks

In this study, we provide causal evidence that improved market access may increase forest loss. In rural Ghana, Twifo Oil Palm Plantation production contracts improved market access in targeted villages by easing credit constraints and reducing transportation and transaction costs, leading to large increases in forest loss following the introduction of the program. In some estimates, annual forest loss more than doubled. This finding is robust across a variety of models and control group specifications.

However, these findings should be interpreted in their appropriate context. First, our study area is geographically narrow—focused on a particular area in the Central Region of Ghana. While this narrow geographic scope minimizes potentially confounding unobserved heterogeneity across regions and countries that might emerge due to factors like ethnic composition, property rights, and ecological differences, it does raise questions of external validity. It is not clear that these results will generalize to other rural agricultural settings, particularly those that differ in land tenure systems. Second, although the dramatic increase in forest loss that we document is concerning due to the ecological damages associated with oil palm monoculture as well as carbon emissions from forest loss, this program likely had

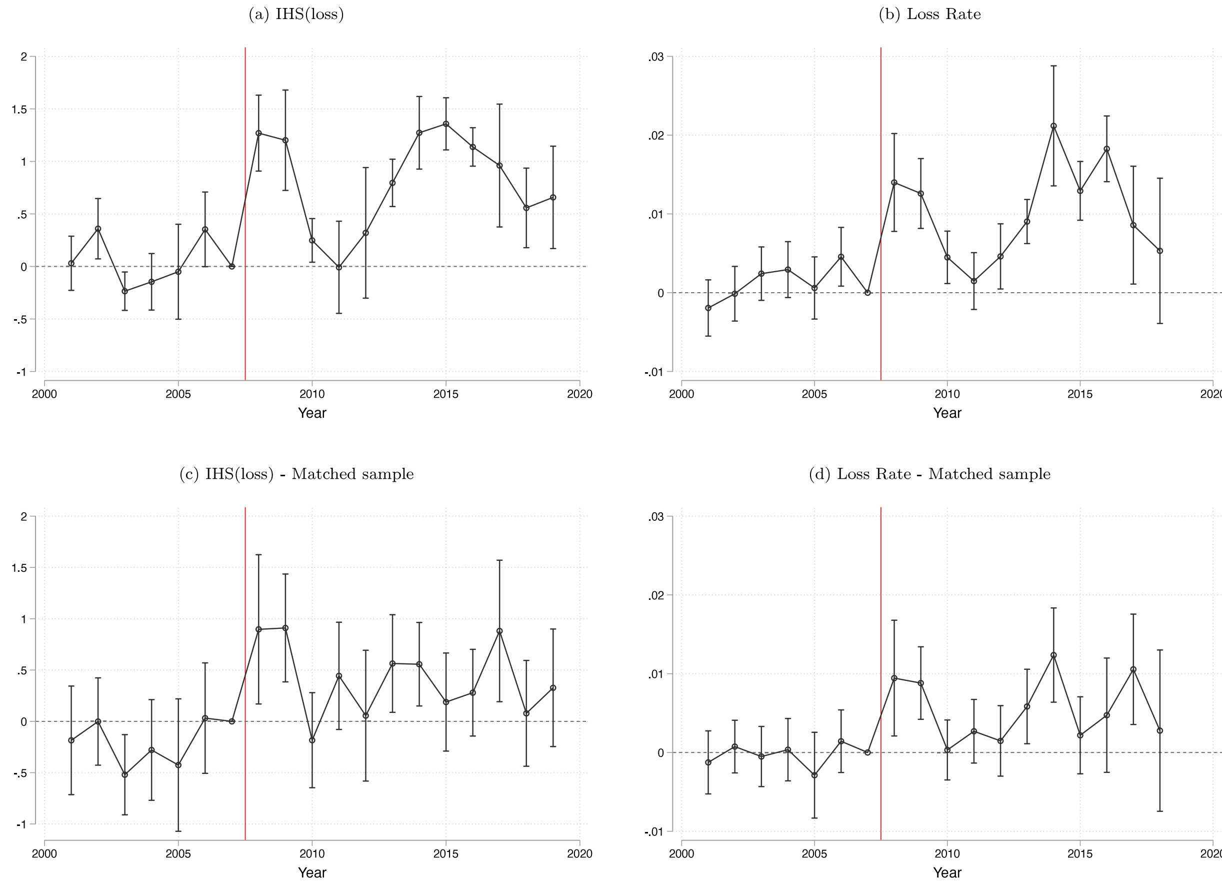

Fig. 2. Event Study Coefficient Estimates.
Notes: These figures present estimates from an event-study comparing annual forest loss in contracting villages to that of other Central-Region villages over time. Differences are normalized to 2007, the year prior to Twifo contracting. Panels (a) and (c) present estimates using the inverse hyperbolic sine of annual forest loss as the outcome variable, while panels (b) and (d) present estimates using the ratio of annual loss to forested area as an outcome. Panels (c) and (d) restrict the control group to matched villages only. All models include village and year fixed effects. Error bars correspond to 95% confidence intervals with Conley standard errors allowing for spatial correlation up to 5 km.

a strong, positive impact on local livelihoods—one that we cannot quantify with the data we have. As a result, we are unable to conduct a full benefit-cost analysis to evaluate the degree to which the benefits of this program were offset by the environmental costs. Finally, while the nature of the program allows us to estimate the causal impacts of the slate of contracting program features, we are unable to disentangle the individual impacts of reductions in transportation costs from the credit access or reduced transaction costs, since all of these changed together. Nevertheless, the limited scope of the program uniquely affects palm oil market access in isolation from all other markets, allowing us to differentiate access shocks to a single agricultural commodity market from other confounding access shocks studied in the literature.

CRediT authorship contribution statement

Ryan Abman: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Clark Lundberg:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jdeveco.2024.103269.

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